

1 Задача

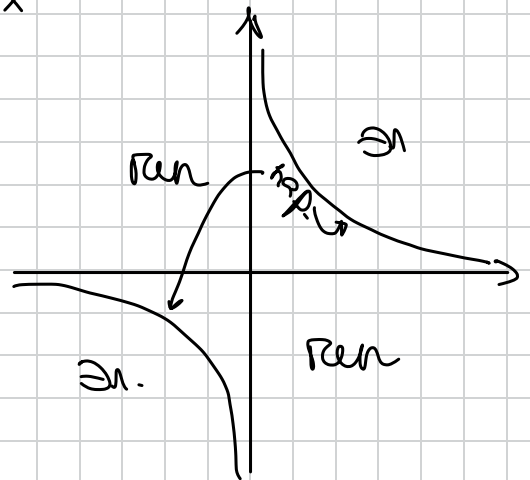
I. Классификация уравнений 2 порядка, характерист.

① а) $yu_{xx} + 2xyu_{xy} + xu_{yy} - u_{xy} = 5x$

$$A = \begin{pmatrix} y & 1 \\ 1 & x \end{pmatrix} \quad \Delta_1 = y$$

$$\Delta_2 = xy - 1$$

элемент. : $\begin{cases} y > 0 \\ y < 0 \\ xy - 1 > 0 \end{cases}$



② $u_{xx} + 2\alpha u_{xz} + u_{yy} + 4\beta u_{yz} + 4u_{zz} = 0$

$$A = \begin{pmatrix} 1 & 0 & \alpha \\ 0 & 1 & 2\beta \\ \alpha & 2\beta & 4 \end{pmatrix} \quad \Delta_1 = 1 > 0$$

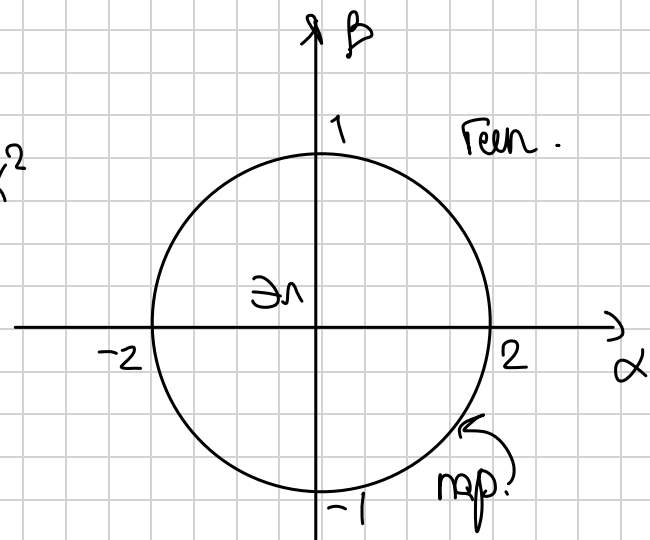
$$\Delta_2 = 1 > 0$$

$$\Delta_3 = 4 - 4\beta^2 - \alpha^2$$

элемент. : $\alpha^2 + 4\beta^2 < 4$

парабол. : $\alpha^2 + 4\beta^2 = 4$

гиперб. : $\alpha^2 + 4\beta^2 > 4$



II. Прив. к кан. виду ур. 2 пор. В случае 2 разг. перем. в ОН.

① б) $x^3 u_{xx} - xy u_{xy} - 2y^2 u_{yy} + x u_x - 2y u_y = 3xy^2$,

1) $x^3 dy^2 + xy dy dx - 2y^2 dx^2 = 0 \Rightarrow$

$$u|_{x=1} = 3e^y$$

$$u_x|_{x=1} = -y^2$$

$$\Rightarrow \begin{cases} y_{x1}' = -2\frac{y}{x} \\ y_{x2}' = \frac{y}{x} \end{cases} \Rightarrow \begin{cases} y = x^{-2}C \\ y = xC \end{cases} \Rightarrow \begin{cases} \xi = yx^2 \\ \eta = \frac{y}{x} \end{cases}$$

$$\begin{cases} u_x = u_\xi \cdot (2xy) - \ln\left(\frac{y}{x^2}\right) \\ u_y = u_\xi(x^2) + \ln\left(\frac{1}{x}\right) \\ u_{xx} = u_{\xi\xi} \cdot 4x^2y^2 + \ln\frac{y^2}{x^4} - 4\frac{y^2}{x} u_{\xi\eta} + u_\xi \cdot 2xy + u_\eta \cdot \frac{2y}{x^3} \\ u_{yy} = u_{\xi\xi}x^4 + \ln\frac{1}{x^2} + 2x u_{\xi\eta} \\ u_{xy} = u_{\xi\xi} \cdot 2x^3y - \ln\frac{y}{x^3} + u_{\xi\eta} \cdot y + u_\xi \cdot 2x - \ln\frac{1}{x^2} \end{cases}$$

$$u_{\xi\xi} \cdot 0 + \ln\eta \cdot 0 + u_{\xi\eta}(-2xy^2) + u_\xi \cdot 0 + \ln\eta \cdot 0 = 9xy^2 \Rightarrow$$

$$\Rightarrow u_{\xi\eta} = -1 \quad u_\xi = \int -1 d\eta + \alpha(\xi) = -\eta + \alpha(\xi)$$

$$u = -\xi\eta + \varphi(\xi) + \kappa(\eta) = \underline{-y^2x + \varphi(yx^2) + \kappa(y/x)}$$

$$\begin{cases} u|_{x=1} = -y^2 + \varphi(y) + \kappa(y) = 3e^y \\ u_x|_{x=1} = -y^2 + \varphi'(y) \cdot 2y + \kappa'(y)(-y) = -y^2 \end{cases} \Rightarrow \varphi(\xi) + \kappa(\eta)$$

$$\Rightarrow -2y + \varphi'(y) + \kappa'(y) = 3e^y \Rightarrow \varphi(y) = e^y + \frac{1}{3}y^2 + C \Rightarrow$$

$$\Rightarrow \kappa(y) = 2e^y + \frac{2}{3}y^2 - C \Rightarrow \underline{u = -y^2x + e^{yx^2} + \frac{1}{3}y^2x^4 + 2e^{y/x} + \frac{2}{3}\frac{y^2}{x^2}}$$

$$\Gamma) y u_{xx} + (x-y) u_{xy} - x u_{yy} - u_x + u_y = 0, \quad u|_{y=0} = 2x^2$$

$$u_y|_{y=0} = 2x$$

$$y(dy)^2 - (x-y)dx dy + x(dx)^2 = 0 \Rightarrow y'_x = \begin{bmatrix} -1 \\ x/y \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} x+y = c_1 \\ y^2 = x^2 + c_2 \end{cases} \Rightarrow \begin{cases} \xi = x+y \\ \eta = y^2 - x^2 \end{cases}$$

$$\begin{cases} u_x = u_\xi + u_\eta(-2x) \\ u_y = u_\xi + u_\eta(2y) \end{cases}$$

$$u_{xx} = u_{\xi\xi} + u_{\eta\eta} 4x^2 + 2u_{\xi\eta} (1 \cdot (-2x)) + u_{\xi} \cdot 0 + u_{\eta} (-2)$$

$$u_{yy} = u_{\xi\xi} + u_{\eta\eta} 4y^2 + 2u_{\xi\eta} \cdot 2y + u_{\xi} \cdot 0 + u_{\eta} \cdot 2$$

$$u_{xy} = u_{\xi\xi} + u_{\eta\eta} (-2x)(2y) + u_{\xi\eta} (2y - 2x)$$

$$y u_{\xi\xi} + 4x^2 y u_{\eta\eta} + 4xy u_{\xi\eta} - 2y u_{\eta} - x u_{\xi\xi} - 4xy^2 u_{\eta\eta} + 4xy u_{\xi\eta} - 2x u_{\eta} +$$

$$+ (x \cdot y) u_{\xi\xi} - 4xy(x-y) u_{\eta\eta} - 2(x-y)^2 u_{\xi\eta} - u_{\xi} + 2x u_{\eta} + u_{\xi} + 2y u_{\eta} = 0 \Rightarrow$$

$$\Rightarrow u_{\xi\eta}(\dots) = 0 \Rightarrow u = f(\xi) + g(\eta) = f(x+y) + g(y^2-x^2)$$

$$\begin{cases} u|_{y=0} = f(x) + g(-x^2) = 2x^2 \\ u_y|_{y=0} = f'(x) + g'(-x^2) \cdot 2 \cdot 0 = 2x \end{cases} \Rightarrow \begin{cases} f(x) = x^2 + C \\ g(t) = -t - C \end{cases}$$

$$\Rightarrow \underline{u = 2x^2 + 2xy}$$

$$g) \quad y^4 u_{yy} + y^2 u_{xy} - 2u_{xx} + 2y^3 y_y = 0, \quad u|_{y=1} = x^2 + 5 \\ u_y|_{y=1} = 2x - 6$$

$$y^4(dx)^2 - y^2 dx dy - 2dy^2 = 0 \Rightarrow$$

$$\Rightarrow dx = \begin{cases} 2 dy/y^2 \\ -dy/y^2 \end{cases} \Rightarrow \begin{cases} \xi = x + \frac{2}{y} \\ \eta = x - \frac{1}{y} \end{cases}$$

$$\begin{cases} u_x = u_{\xi} + u_{\eta} \\ u_y = u_{\xi} \left(-\frac{2}{y^2}\right) + u_{\eta} \left(\frac{1}{y^2}\right) \end{cases}$$

$$u_{xx} = u_{\xi\xi} \cdot 1 + u_{\eta\eta} \cdot 1 + 2u_{\xi\eta} \cdot 1 \cdot 1$$

$$u_{yy} = u_{\xi\xi} \left(\frac{4}{y^4}\right) + u_{\eta\eta} \left(\frac{1}{y^4}\right) + 2u_{\xi\eta} \left(-\frac{2}{y^2} \cdot \frac{1}{y^2}\right) + u_{\xi} \frac{4}{y^3} + u_{\eta} \left(-\frac{2}{y^3}\right)$$

$$u_{xy} = u_{\xi\xi} \left(-\frac{2}{y^2}\right) + u_{\eta\xi} \left(-\frac{1}{y^2}\right) + u_{\eta\eta} \left(\frac{1}{y^2}\right)$$

$$u_{yy}(-2+4-2) + u_{yy}(-2+1+1) + u_{yy}(-4-4-1) + u_{yy}(-4y+4y) + u_{yy}(2y-2y) = 0$$

$$\Rightarrow u_{yy} = 0 \Rightarrow u = f(\xi) + g(\eta) = f(x + \frac{2}{y}) + g(x - \frac{1}{y})$$

$$\begin{cases} u|_{y=1} = f(x+2) + g(x-1) = x^2 + 5 \\ u_y|_{y=1} = f'(\frac{2}{1}) + g'(\frac{1}{1}) = 2x - 6 \end{cases} \Rightarrow \begin{cases} f'(x+2) = 2 \\ g'(x-1) = 2x - 2 \end{cases} \Rightarrow$$

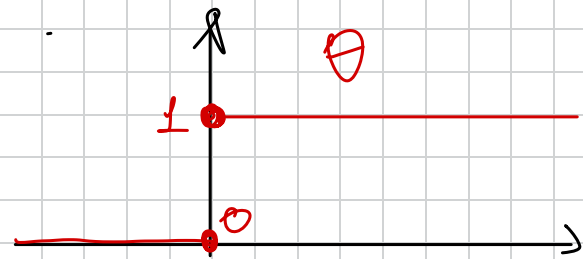
$$\Rightarrow \begin{cases} f'(t) = 2 \\ g'(t) = 2t \end{cases} \Rightarrow \begin{cases} f(t) = 2t + C \\ g(t) = t^2 - C \end{cases} \Rightarrow u = 2x + \frac{4}{y} + x^2 - \frac{2x}{y} + \frac{1}{y^2}$$

III. Задача Даламбера

② а) $u_{tt} = a^2 u_{xx}$, $x \in \mathbb{R}^1$, $t > 0$

$$u|_{t=0} = \varphi(x) \quad u_t|_{t=0} = 0$$

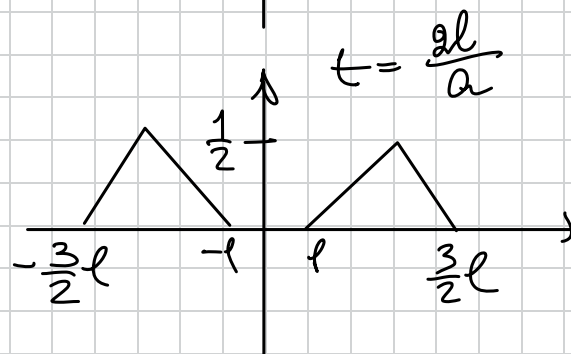
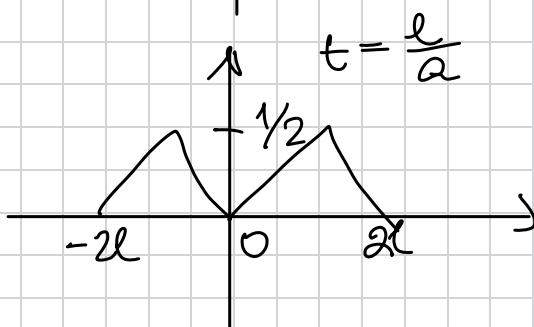
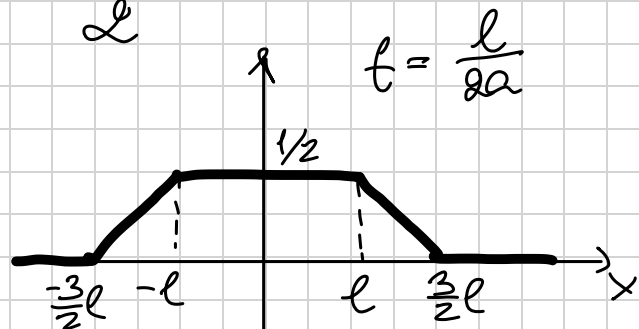
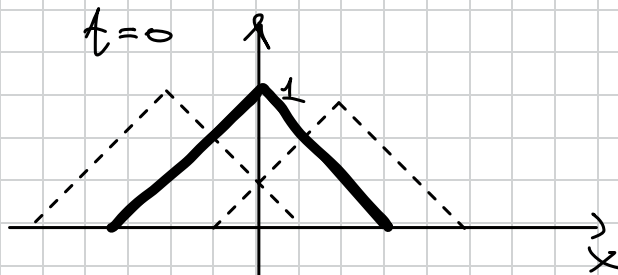
$$\varphi(x) = (1 - \frac{|x|}{l}) \Theta(l - |x|)$$



$$l = \text{const} > 0$$

График $t = \frac{l}{2a}$; $\frac{l}{a}$; $\frac{2l}{a}$

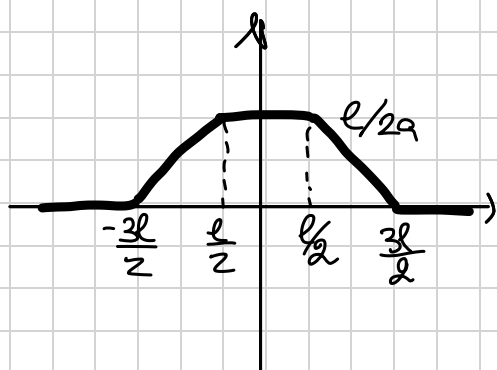
Решение: $u(x, t) = \frac{\varphi(x+at) + \varphi(x-at)}{2}$



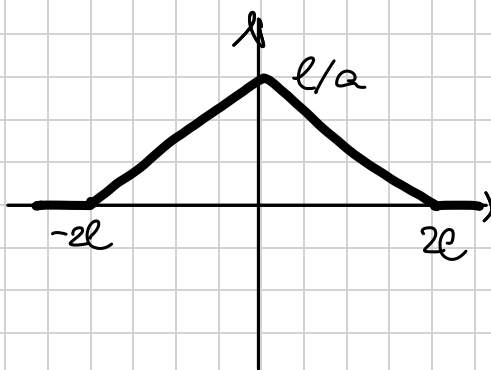
$$\delta) \begin{cases} u_{tt} = a^2 u_{xx} \\ u(t=0) = 0, \quad u_t(t=0) = \psi(x) \end{cases} \quad \psi(x) = \Theta(l - |x|)$$

$$u(x, t) = \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi = \frac{1}{2a} \int_{x-at}^{x+at} \Theta(l - |\xi|) d\xi$$

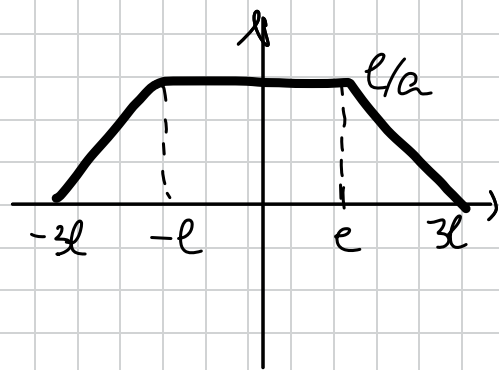
1) $t = \frac{l}{2a}$



2) $t = \frac{l}{a}$



3) $t = \frac{2l}{a}$



21.2. Сформулируйте, что такое $\begin{cases} u_{tt} = a^2 u_{xx}, \quad t > 0, \quad x > 0 \\ u(t=0) = u_0(x) \quad u_t(t=0) = u_1(x) \\ u(x=0) = 0 \end{cases}$

ли. эк. прел. $u(x, t) = \begin{cases} \frac{u_0(x+at) + u_0(x-at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} u_1(\xi) d\xi & x \geq at \\ \frac{u_0(x+at) - u_0(x-at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} u_1(\xi) d\xi & x < at \end{cases}$

□ Проформулируйте кр. экрзом u_0 и u_1 , змешен

что узн. Вспомогательные: В случае $x \geq at$ кр. экр. не

узн., при $0 \leq x < at$: $\frac{\tilde{u}_0(x+at) + \tilde{u}_0(x-at)}{2} = \frac{u_0(x+at) - u_0(at-x)}{2}$

$$\int_{x-at}^{x+at} \tilde{u}_1(\xi) d\xi = \int_{x-at}^0 u_1(\xi) d\xi + \int_0^{x+at} u_1(\xi) d\xi = \int_{at-x}^{x+at} u_1(\xi) d\xi$$

21.19. Найти $\exists!$ прел. $\begin{cases} u_{tt} = 3u_{xx} + 2(1-t^2)e^{-2x} \\ u(t=0) = 1, \quad u_t(t=0) = x, \quad (u_x - 2u)|_{x=0} = \end{cases}$

Lösung z.p. $f(t)e^{-2x}$: $f''e^{-2x} = 3f \cdot 4e^{-2x} + 2(1-6t^2)e^{-2x} \Rightarrow$
 $\Rightarrow f'' - 12f = 2 - 12t^2 \Rightarrow f = t^2 \Rightarrow$
 $\Rightarrow$ andere partikuläre $fe^{-2x} = t^2e^{-2x}$

Hypoth $w = u - fe^{-2x}$: $w_{tt} = 3w_{xx}$

$w|_{t=0} = 1 - (C_1 + C_2)e^{-2x}$ $w_t|_{t=0} = x - 2\sqrt{3}(C_1 - C_2)e^{-2x}$

$(w_x - 2w)|_{x=0} = -2 + t - 4t^2 + 2f + 2f = -2 + t$

$w(x, t) = f(x + \sqrt{3}t) + g(x - \sqrt{3}t)$

- Lösung partikuläre & allg. $x > t$:

$$\begin{cases} w|_{t=0} = f(x) + g(x) = 1 \\ w_t|_{t=0} = \sqrt{3}f'(x) - \sqrt{3}g'(x) = x \end{cases} \Rightarrow \begin{cases} f'(x) = \frac{x}{2\sqrt{3}} \\ g'(x) = \frac{-x}{2\sqrt{3}} \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} f(x) = \frac{x^2}{4\sqrt{3}} + \tilde{C}_1 \\ g(x) = \frac{-x^2}{4\sqrt{3}} + \tilde{C}_2 \end{cases} \Rightarrow \tilde{C}_1 + \tilde{C}_2 = 1$$

$w = \frac{1}{4\sqrt{3}}(x + \sqrt{3}t)^2 - \frac{1}{4\sqrt{3}}(x - \sqrt{3}t)^2 + \tilde{C}_1 + \tilde{C}_2 = \underline{\underline{xt + 1}}$

- Lösung part. unter $x < t$, $f(t)$ -quere.

$(w_x - 2w)|_{x=0} = -2 + t \Rightarrow -2 + t = f'(t) + g'(-t) - 2f(t) - 2g(-t) \Rightarrow$

$\Rightarrow \frac{t}{2\sqrt{3}} - \frac{t^2}{2\sqrt{3}} - 2\tilde{C}_1 + g'(-t) - 2g(-t) = -2 + t$

Hypoth $p = -t \Rightarrow g'(p) - 2g(p) = -2 - p + 2\tilde{C}_1 + \frac{p^2}{2\sqrt{3}} + \frac{p}{2\sqrt{3}} \Rightarrow$

$\Rightarrow g_2(p) = Ap^3 + Bp^2 + Cp + D \Rightarrow$

$$\Rightarrow \cancel{3Ap^3} + 2Bp + C - \cancel{2Ap^3} - 2Bp^2 - 2Cp - 2D = -2 + 2\tilde{C}_1 + p(\frac{1}{2\sqrt{3}} - 1)$$

$$+ \frac{p^2}{2\sqrt{3}} \Rightarrow A=0, B=\frac{-1}{4\sqrt{3}}, C=1-\frac{1}{2\sqrt{3}}, D=\underbrace{+\frac{3}{2}-\tilde{C}_1}_{\frac{1}{2}+\tilde{C}_2}-\frac{1}{4\sqrt{3}}$$

$$g(p) = \tilde{C}e^{2p} - \frac{1}{4\sqrt{3}}p^2 + (1-\frac{1}{2\sqrt{3}})p + \tilde{C}_2 + \frac{1}{4\sqrt{3}} - \frac{1}{2}$$

$$\text{Умова: } g(+0) = g(-0) \Rightarrow \tilde{C}_2 = \tilde{C}_2 + \frac{1}{4\sqrt{3}} - \frac{1}{2} + \tilde{C} \Rightarrow \tilde{C} = \frac{1}{2} - \frac{1}{4\sqrt{3}}$$

$$\text{Нам } u(x,t) = t^2 e^{-2x} + \frac{(x+\sqrt{3}t)^2}{4\sqrt{3}} + 1 + \int \frac{-(x-\sqrt{3}t)^2}{4\sqrt{3}}, x \geq t \\ \left(\frac{1}{2} - \frac{1}{4\sqrt{3}} \right) e^{2(x-\sqrt{3}t)} - \\ - \left(1 - \frac{1}{2\sqrt{3}} \right) (x-\sqrt{3}t) - \frac{1}{4\sqrt{3}} (x-\sqrt{3}t)^2 + \frac{1}{4\sqrt{3}} - \frac{1}{2}$$

$$\textcircled{3} \text{ B) } \begin{cases} 4u_{tt} = u_{xx} - 3\sin(x+t) \\ u|_{t=0} = \sin x + \sin 2x, u_t|_{t=0} = 2 + \cos x \\ (u_x - 2u)|_{x=0} = 2 - 4t - \sin t + \cos t \end{cases}$$

$$\text{Загальне перешеве: } u_2 = \sin(x+t)$$

$$\text{Нехай } u = v + u_2 : v_{tt} = \frac{1}{4}v_{xx}, v|_{t=0} = \sin 2x, v_t|_{t=0} = 2 \\ (v_x - 2v)|_{x=0} = 2 - 4t + \sin t$$

$$v = f(x + \frac{1}{2}t) + g(x - \frac{1}{2}t)$$

$$\text{Нехай перешеве в області } x \geq \frac{1}{2}t$$

$$\begin{cases} v|_{t=0} = f(x) + g(x) = \sin 2x \\ v_t|_{t=0} = \frac{1}{2}f'(x) - \frac{1}{2}g'(x) = 2 \end{cases} \Rightarrow \begin{cases} f'(x) = 2 + 2\cos 2x \Rightarrow \\ \Rightarrow \int f(x) = 2x + \frac{1}{2}\sin 2x + C \\ g(x) = -2x + \frac{1}{2}\sin 2x - C \end{cases}$$

$$\text{Нехай в області } x < \frac{1}{2}t$$

$$(v_x - 2v)|_{x=0} = f'(\frac{1}{2}t) + g'(\frac{1}{2}t) - 2f(\frac{1}{2}t) - 2g(-\frac{1}{2}t) = 2 - 4t + \sin t$$

$$g'(-\frac{1}{2}t) - 2p(-\frac{1}{2}t) = -2t + 2\sin t - \cos t + 2t, \text{ allora } p = -\frac{1}{2}t$$

$$g'(p) - 2p(p) = 4p + 2\sin(-2p) - \cos(-2p) + 2t$$

$$g_1 = \alpha p + \beta \quad \alpha - 2\alpha p - 2\beta = 4p + 2C \Rightarrow g_1 = -2p - 1 - C$$

$$g_2 = \alpha \sin(-2p) + \beta \cos(-2p) \Rightarrow -2\alpha \cos(-2p) + 2p \sin(-2p) - 2\alpha \sin(-2p)$$

$$- 2\beta \cos(-2p) = 2\sin(-2p) - \cos(-2p) \Rightarrow \begin{cases} \alpha = -1/4 \\ \beta = 3/4 \end{cases} \Rightarrow$$

$$\Rightarrow g_2 = \frac{3}{4} \cos(-2p) - \frac{1}{4} \sin(-2p) \Rightarrow$$

$$\Rightarrow g(p) = \tilde{C} e^{2p} - 2p - 1 - C + \frac{3}{4} \cos(-2p) - \frac{1}{4} \sin(-2p)$$

$$\text{Condizione: } g(+0) = g(-0) \Rightarrow \cancel{C} = \tilde{C} - 1 - \cancel{C} + 3/4 \Rightarrow \tilde{C} = \frac{1}{4} \quad x > \frac{t}{2}$$

$$\text{Verifica } u = \sin(x+t) + 2(x+\frac{t}{2}) + \frac{1}{2} \sin 2(x+\frac{t}{2}) + \begin{cases} -2(x-\frac{t}{2}) + \frac{1}{2} \sin 2(x-\frac{t}{2}) \\ \frac{1}{4} e^{2(x-\frac{t}{2})} - 2(x-\frac{t}{2}) - 1 + \frac{3}{4} \cos(-2(x-\frac{t}{2})) - \frac{1}{4} \sin(-2(x-\frac{t}{2})) \end{cases}$$

II Задача

12.43. 5)
$$\begin{cases} u_{tt} = a^2 \Delta u \\ u|_{t=0} = 2x^2 - y^2 = P(x, y) \\ u_t|_{t=0} = 2x^2 + y^2 = Q(x, y) \end{cases}$$

$$u = u|_{t=0} + t u_t|_{t=0} + \frac{t^2}{2} u_{tt}|_{t=0} + \dots$$

$$u_{tt}|_{t=0} = a^2 \Delta u|_{t=0} = a^2 \Delta P$$

$$u_{ttt}|_{t=0} = a^2 \Delta(u_t|_{t=0}) = a^2 \Delta Q$$

$$u_{tttt}|_{t=0} = a^2 \Delta(a^2 \Delta P) = 0$$

$$\begin{aligned} u &= 2x^2 - y^2 + t(2x^2 + y^2) + \frac{t^2}{2} \cdot 2(4-2) + \frac{t^3}{6} 2(4+2) = \\ &= 2x^2 - y^2 + t(2x^2 + y^2) + 2t^2 + 2t^3 \end{aligned}$$

12.44. 7)
$$\begin{cases} u_{tt} = a^2 \Delta u + \cos x \sin y e^z - \text{c.p.} \\ u|_{t=0} = x^2 e^{y+z} - \text{норм c.p.} \\ u_t|_{t=0} = \sin x e^{y+z} - \text{c.p.} \end{cases}$$

Поиск $u = v + w + p$

1)
$$\begin{cases} v_{tt} = a^2 \Delta v + \cos x \sin y e^z \\ v|_{t=0} = 0 \\ v_t|_{t=0} = 0 \end{cases}$$

2)
$$\begin{cases} w_{tt} = a^2 \Delta w \\ w|_{t=0} = x^2 e^{y+z} \\ w_t|_{t=0} = 0 \end{cases}$$

3)
$$\begin{cases} p_{tt} = a^2 \Delta p \\ p|_{t=0} = 0 \\ p_t|_{t=0} = \sin x e^{y+z} \end{cases}$$

$$1) \Delta(\cos x \sin y e^z) = -\cos x \sin y e^z$$

$$V = h(t) \cos x \sin y e^z \Rightarrow h'' \cos x \sin y e^z = a^2 h (-1) \cos x \sin y e^z + \cos x \sin y e^z$$

$$\begin{cases} h'' = -a^2 h + 1 \\ h(0) = 0 \\ h'(0) = 0 \end{cases}$$

$$h(t) = A \sin at + B \cos at + C \Rightarrow$$

$$\Rightarrow h''(t) = a^2 A \sin at - a^2 B \cos at = C = \frac{1}{a^2}$$

$$= -a^2 h + 1 = -a^2 (A \sin at + B \cos at) - a^2 C + 1 \Rightarrow$$

$$h(0) = B + C = 0 \Rightarrow B = -C = -\frac{1}{a^2}, \quad h'(0) = Aa = 0 \Rightarrow A = 0$$

$$h(t) = \frac{1}{a^2} (1 - \cos at) \Rightarrow V = \frac{1}{a^2} (1 - \cos at) \cos x \sin y e^z$$

$$2) \Delta(x^2 e^{y+z}) = \underbrace{2e^{y+z}}_{g.-c.p.} + \underbrace{2x^2 e^{y+z}}_{\Delta f}$$

$$W = h_1(t) x^2 e^{y+z} + h_2(t) e^{y+z}$$

$$h_1'' x^2 e^{y+z} + h_2'' e^{y+z} = a^2 h_1 (2e^{y+z} + 2x^2 e^{y+z}) + a^2 h_2 (2e^{y+z}) \Rightarrow$$

$$\Rightarrow \begin{cases} h_1'' = 2h_1 a^2 \\ h_1(0) = 1 \\ h_1'(0) = 0 \end{cases} \quad \begin{cases} h_2'' = (2h_1 + 2h_2) a^2 \\ h_2(0) = 0 \\ h_2'(0) = 0 \end{cases}$$

$$h_1 = A \cosh \sqrt{2} at + B \sinh \sqrt{2} at = \cosh \sqrt{2} at$$

$$h_1(0) = B = 1$$

$$h_1'(0) = Aa = 0 \Rightarrow A = 0$$

$$h_2'' = 2a^2 h_2 + 2a^2 \cosh \sqrt{2} at - \text{это переписать, нужен ответ}$$

$$h_{2e} = A t \cosh \sqrt{2} at + B t \sinh \sqrt{2} at \Rightarrow$$

$$\Rightarrow h_{2e}'' = 2A \sqrt{2} a \cosh \sqrt{2} at + A t 2a^2 \sinh \sqrt{2} at + 2B \sqrt{2} a \sinh \sqrt{2} at + B t 2a^2 \cosh \sqrt{2} at = 2a^2 A t \sinh \sqrt{2} at + 2a^2 B t \cosh \sqrt{2} at +$$

$$+ 2a^2 \cosh \sqrt{2}at \Rightarrow \begin{cases} A = \frac{2}{\sqrt{2}} \\ B = 0 \end{cases} \Rightarrow h_{22} = \frac{a}{\sqrt{2}} \cosh \sqrt{2}at = h_2$$

$$h_{22y} = 0 \text{ uz atdarysimas } h_1 \text{ uostem}$$

Tarp $w = \cosh \sqrt{2}at x^2 e^{y+z} + \frac{a}{\sqrt{2}} \cosh \sqrt{2}at e^{y+z}$

$$3) \Delta(\sin x e^{y+z}) = \sin x e^{y+z} \Rightarrow \text{c.p.} \Rightarrow p = h(t) \sin x e^{y+z}$$

$$h'' \sin x e^{y+z} = a^2 \sin x e^{y+z} \cdot h \Rightarrow h'' = a^2 h$$

$$\begin{cases} h(t) = A \cosh at + B \sinh at = \frac{1}{a} \sinh at \\ h(0) = 0 = B \end{cases}$$

$$h'(0) = 1 = Aa \Rightarrow A = 1/a$$

$$\Rightarrow p = \frac{1}{a} \sinh at + \sin x e^{y+z}$$

$$\text{tada } u = v + w + p$$

4) g) $\begin{cases} u_{tt} = \frac{1}{5} \Delta u + 2t^2 \cos(x+2y) - \text{c.p.} \\ u|_{t=0} = yz^3 - \text{atdarysimas.} \\ u_t|_{t=0} = \frac{1}{1+(x+2y)^2} - \text{nere. randa.} \end{cases}$

$$\text{tada } u = v + w + p$$

$$1) \begin{cases} v_{tt} = \frac{1}{5} \Delta v + 2t^2 \cos(x+2y) \\ v|_{t=0} = 0 \\ v_t|_{t=0} = 0 \end{cases}$$

$$2) \begin{cases} w_{tt} = \frac{1}{5} \Delta w \\ w|_{t=0} = yz^3 \\ w_t|_{t=0} = 0 \end{cases}$$

$$3) \begin{cases} p_{tt} = \frac{1}{5} \Delta p \\ p|_{t=0} = 0 \\ p_t|_{t=0} = \frac{1}{1+(x+2y)^2} \end{cases}$$

$$1) v = h(t) \cos(x+2y)$$

$$h'' \cos(x+2y) = \frac{1}{5} h(-5) \cos(x+2y) + 2t^2 \cos(x+2y)$$

$$\begin{cases} h'' = -h + 2t^2 \\ h(0) = 0 \\ h'(0) = 0 \end{cases}$$

$$h_2 = \alpha t^2 + \beta t + \gamma$$

$$2\alpha = -\alpha t^2 - \beta t - \gamma + 2t^2 \Rightarrow \alpha = 2, \beta = 0, \gamma = -4 \Rightarrow$$

$$\Rightarrow \begin{cases} h = 2t^2 - 4 + A \sin t + B \cos t \\ h(0) = -4 + B = 0 \\ h'(0) = A = 0 \end{cases} \Rightarrow \begin{cases} B = 4 \\ A = 0 \end{cases} \Rightarrow h(t) = 2t^2 - 4 + 4 \cos t \Rightarrow$$

$$\Rightarrow \underline{V = (2t^2 - 4 + 4 \cos t) \cos(x + 2y)}$$

$$2) \quad W = \underbrace{W|_{t=0}}_{y z^3} + \cancel{t W|_{t=0}} + \frac{t^2}{2} \underbrace{W''|_{t=0}}_{\frac{1}{5}(6yz)} + \dots \Rightarrow$$

$$\Rightarrow \underline{W = y z^3 + \frac{t^2}{2} \cdot \frac{6}{5} y z}$$

$$3) \quad p = p(t, x - 2z) = p(t, \xi) \Rightarrow \begin{cases} p_{tt} = \frac{1}{5} p_{\xi\xi} \cdot 5 = p_{\xi\xi} \\ p|_{t=0} = 0 \\ p_t|_{t=0} = \frac{1}{1+\xi^2} \end{cases}$$

$$p = \frac{1}{2 \cdot 1} \int_{\xi-t}^{\xi+t} \frac{1}{1+\eta^2} d\eta = \frac{1}{2} (\arctan(\xi+t) - \arctan(\xi-t)) \Rightarrow$$

$$\Rightarrow p = \frac{1}{2} [\arctan((x-2z)+t) - \arctan((x-2z)-t)]$$

$$\text{Also } u = v + w + p$$

$$(5) \quad \begin{cases} u_{tt} = a^2 \Delta u, \quad t > 0, x \in \mathbb{R}^3 \\ u|_{t=0} = u_0(x) = \alpha(r), \quad u_0 \in C^3(\mathbb{R}^3) \\ u_t|_{t=0} = u_1(x) = \beta(r), \quad u_1 \in C^2(\mathbb{R}^3) \end{cases}$$

$$u = u(t, \sqrt{x^2 + y^2 + z^2}) = u(t, r) \quad u(t, r) = \frac{U(r, t)}{r}$$

$$\begin{cases} U_{tt} = a^2 U_{rr} \\ U|_{t=0} = r u_0(r) = r \alpha(r) \quad + \text{homogen. cond. for } r=0 \\ U_t|_{t=0} = r \beta(r) \\ U|_{r=0} \end{cases}$$

no q-ne ~~transfere~~ $V(t,r) = \frac{\alpha(r+at) + \alpha(r-at)}{2} + \frac{1}{2a} \int_{r-at}^{r+at} \beta(\xi) d\xi$

$$u(t,r) = \frac{V(t,r)}{r} = \frac{\alpha(r+at) + \alpha(r-at)}{r} + \frac{1}{2ar} \int_{r-at}^{r+at} \beta(\xi) d\xi$$

⑥
$$\begin{cases} u_t = \Delta u \\ u|_{t=0} = 0 \\ u|_{t=0} = \int \ln(1+x^2+y^2+z^2), (x,y,z) \in \mathbb{B} \\ 0, (x,y,z) \in \mathbb{B} \end{cases} \quad \mathbb{B} = \{y > 0, 0 < x < \pi\}$$

$$u(t,0,0,0) = \frac{1}{4\pi t} \int_{|\xi|=at \cap \mathbb{B}} \ln(1+\xi_1^2 + \xi_2^2 + \xi_3^2) dS_\xi = \frac{1}{t} \ln(1+a^2 t^2) \int d\xi =$$

$$= \frac{1}{t} \ln(1+a^2 t^2) \cdot \frac{1}{6} 4\pi a^2 t^2 = \frac{\pi t}{6} \ln(1+\frac{t^2}{4})$$

~~~~~ Aufgabe 2 (exponential) ~~~~~

① a) 
$$\begin{cases} u_t = \Delta u + e^{-9t} \cos(2x-y+2z), t > 0 \\ u|_{t=0} = xy^2z^3 + \cos y \sin(x+y+z) \end{cases}$$

1) 
$$\begin{cases} v_t = \Delta v + e^{-9t} \cos(2x-y+2z) \\ v|_{t=0} = 0 \end{cases}$$

2) 
$$\begin{cases} w_t = \Delta w \\ w|_{t=0} = xy^2z^3 \end{cases}$$

3) 
$$\begin{cases} p_t = \Delta p \\ p|_{t=0} = \cos y \sin(x+y+z) \end{cases}$$

1) 
$$\Delta(\cos(2x-y+2z)) = (-4-1-4)\cos(2x-y+2z)$$

$$V = h(t) \cos(2x-y+2z)$$

$$h' \cos(2x-y+2z) = h(-9) \cos(2x-y+2z) + e^{-9t} \cos(2x-y+2z)$$

$$\begin{cases} h' + gh = e^{-gt} \\ h(0) = 0 \end{cases} \Rightarrow \begin{cases} h = (At+B)e^{-gt} \\ Ae^{-gt} + At(-g)e^{-gt} - gBe^{-gt} + gAt e^{-gt} + Be^{-gt} = e^{-gt} \end{cases}$$

$$\begin{cases} A - 8B = 1 \\ B = 0 \end{cases} \Rightarrow \begin{cases} A = 1 \\ B = 0 \end{cases}$$

$$\underline{V = e^{-gt} \cos(2x - y + 2z)}$$

$$2) \begin{cases} W_t = \delta W \\ W|_{t=0} = xy^2z^3 = P \end{cases}$$

$$W = \underbrace{W|_{t=0}}_P + t \underbrace{W_t|_{t=0}}_{\Delta P} + \frac{t^2}{2} \underbrace{W_{tt}|_{t=0}}_{\Delta(\Delta P)} + \dots$$

$$\Delta P = 2xz^3 + 6xy^2z$$

$$\Delta(\Delta P) = 12xz + 12xz = 24xz \quad \Delta(\Delta(\Delta P)) = 0$$

$$\underline{W = xy^2z^3 + t(2xz^3 + 6xy^2z) + \frac{t^2}{2} 24xz}$$

$$3) \cos \sin(x+y+z) = \frac{1}{2} \sin(x+2y+z) + \frac{1}{2} \sin(x+z)$$

$$\Delta \sin(x+2y+z) = (-1-4-1) \sin(x+2y+z)$$

$$\Delta \sin(x+z) = -2 \sin(x+z)$$

$$P = h_1(t) \sin(x+2y+z) + h_2(t) \sin(x+z)$$

$$h_1' \sin(x+2y+z) + h_2' \sin(x+z) = h_1(-6) \sin(x+2y+z) + h_2(-2) \sin(x+z)$$

$$\begin{cases} h_1' - 6h_1 = 0 \\ h_1(0) = 1/2 \end{cases} \quad \begin{cases} h_2' + 2h_2 = 0 \\ h_2(0) = 1/2 \end{cases} \Rightarrow h_1 = \frac{1}{2} e^{-6t} \quad h_2 = \frac{1}{2} e^{-2t}$$

$$\underline{P = \frac{1}{2} e^{-6t} \sin(x+2y+z) + \frac{1}{2} e^{-2t} \sin(x+z)}$$

$$\boxed{u = V + W + P}$$

$$g) \begin{cases} u_t = \frac{1}{4}u + t^4(x+1) \\ u|_{t=0} = e^{2x-z^2} \sin(x+y) \end{cases} \quad u = v + w(t, z) \cdot p(t, x, y)$$

$$1) \quad v = h(t)(x+1) \Rightarrow \begin{cases} h' = t^4 \\ h(0) = 0 \end{cases} \Rightarrow h = \frac{t^5}{5} \Rightarrow \\ \Rightarrow w'(x+1) = \frac{1}{4}h \cdot 0 + t^4(x+1) \Rightarrow \Rightarrow \underline{\underline{v = \frac{t^5}{5}(x+1)}}$$

$$2) \quad w(t, z) = \frac{1}{2 \cdot \frac{1}{2} \sqrt{t}} \int_{-\infty}^{+\infty} e^{-\frac{(z-\xi)^2}{\frac{1}{4}t}} e^{2\xi - \xi^2} d\xi \quad \textcircled{=}$$

$$\begin{aligned} & -\frac{\xi^2}{t} + 2\frac{\xi z}{t} - \frac{z}{t} + 2\xi - \xi^2 = -\xi^2\left(\frac{1}{t}+1\right) + 2\xi\left(\frac{z}{t}+1\right) - \frac{z^2}{t} = \\ & = -\left[\left(\xi\sqrt{\frac{1}{t}+1}\right) - 2\xi\sqrt{\frac{1}{t}+1} \cdot \frac{z/t+1}{\sqrt{\frac{1}{t}+1}} + \frac{\left(\frac{z}{t}+1\right)^2}{\frac{1}{t}+1}\right] + \frac{(z+t)^2}{t(t+1)} - \frac{z^2}{t} = \\ & = -\left(\xi\sqrt{\frac{1}{t}+1} - \frac{z+t}{\sqrt{t(t+1)}}\right)^2 + \frac{z^2 + 2zt + t^2 - tz^2 - z^2}{t(t+1)} \end{aligned}$$

$$\begin{aligned} \textcircled{=} & \frac{e^{\frac{2z - z^2 + t}{t+1}}}{\sqrt{t}} \int_{-\infty}^{+\infty} e^{-\left(\xi\sqrt{\frac{1}{t}+1} - \frac{z+t}{\sqrt{t(t+1)}}\right)^2} d\xi = \\ & = \frac{e^{\frac{2z - z^2 + t}{t+1}}}{\sqrt{t}} \frac{1}{\sqrt{\frac{1}{t}+1}} \int_{-\infty}^{+\infty} e^{-s^2} ds = e^{\frac{2z - z^2 + t}{t+1}} \end{aligned}$$

$$h) \begin{cases} u_t = u + x^2 y^2 \sin t \\ u|_{t=0} = xy z \cos x \end{cases} \quad u = v + w(t, x) \cdot p(t, y, z)$$

$$1) \quad \Delta(x^2 y^2) = y^2 + x^2, \quad \Delta(x^2 + y^2) = 4$$

$$v = h_1(t)x^2 y^2 + h_2(t)(x^2 + y^2) + h_3(t) \cdot 1$$

$$h_1' x^2 y^2 + h_2'(x^2 + y^2) + h_3' = h_1(x^2 + y^2) + h_2 \cdot 4 + h_3 \cdot 0 + x^2 y^2 \sin t$$

$$\begin{cases} h_1' = \sin t \\ h_1(0) = 0 \end{cases} \quad \begin{cases} h_2' = h_1 \\ h_2(0) = 0 \end{cases} \quad \begin{cases} h_3' = 4h_2 \\ h_3(0) = 0 \end{cases} \Rightarrow \begin{cases} h_1 = 1 - \cos t \\ h_2 = t - \sin t \\ h_3 = 2t^2 + 4\cos t - 4 \end{cases}$$

$$\underline{V = (1 - \cos t)x^2y^2 + (t - \sin t)(x^2 + y^2) + 2t^2 + 4\cos t - 4}$$

$$2) \quad \begin{cases} w_t = xw \\ w|_{t=0} = x \cos x \end{cases} \quad \begin{cases} p_t = xp \\ p|_{t=0} = yz \end{cases}$$

$$\Delta(x \cos x) = \frac{\partial}{\partial x} (\cos x - x \sin x) = -2 \sin x - x \cos x$$

$$w = h_1(t)x \cos x + h_2(t) \sin x$$

$$h_1' x \cos x + h_2' \sin x = h_1 (-2 \sin x - x \cos x) + h_2 (-1) \sin x$$

$$\begin{cases} h_1' + h_2 = 0 \\ h_1(0) = 1 \end{cases} \quad \begin{cases} h_2' + h_2 = -2h_1 \\ h_2(0) = 0 \end{cases} \Rightarrow \begin{cases} h_1 = e^{-t} \\ h_2 = -2te^{-t} \end{cases}$$

$$\underline{w = e^{-t} x \cos x - 2te^{-t} \cos x}$$

$$\begin{cases} p_t = xp \\ p|_{t=0} = yz \end{cases} \quad p = p|_{t=0} + t p_t|_{t=0} + \dots = \underline{yz}$$

$$\text{also } u = v + w \cdot p$$

$$\underline{\text{B.S. 5) } \begin{cases} u_t = u_{xx} + \sin t \\ u|_{t=0} = e^{-x^2} \end{cases} \Rightarrow u = 1 - \cos t + v : \begin{cases} v_t = v_{xx} \\ v|_{t=0} = e^{-x^2} \end{cases}$$

$$\begin{cases} v_t = v_{xx} \\ v|_{t=0} = e^{-x^2} \end{cases} \quad v(t, x) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{(x-\xi)^2}{4t}} \cdot e^{-\xi^2} d\xi \quad \textcircled{=}$$

$$\frac{1}{4t} \left( -\int (1+4t) - 2\int \sqrt{1+4t} \frac{x}{\sqrt{1+4t}} + \frac{x^2}{\sqrt{1+4t}} + \frac{4tx^2}{1+4t} \right) =$$

$$= \frac{-\left(\xi\sqrt{1+4t} - \frac{x}{\sqrt{1+4t}}\right)^2}{4t} - \frac{x^2}{1+4t}$$

$$\textcircled{=} \frac{e^{\frac{-x^2}{1+4t}}}{2\sqrt{t}} \int_{-\infty}^{+\infty} e^{-\left(\frac{\xi\sqrt{1+4t} - \frac{x}{\sqrt{1+4t}}}{2\sqrt{t}}\right)^2} d\xi = \begin{cases} S = \frac{1}{2\sqrt{t}}\left(\xi\sqrt{1+4t} - \frac{x}{\sqrt{1+4t}}\right) \\ ds = \frac{\sqrt{1+4t}}{2\sqrt{t}} d\xi \end{cases}$$

$$= \frac{e^{\frac{-x^2}{1+4t}}}{2\sqrt{t}} \cdot \frac{2\sqrt{t}}{\sqrt{1+4t}} \int_{-\infty}^{+\infty} e^{-s^2} ds = \frac{e^{\frac{-x^2}{1+4t}}}{\sqrt{1+4t}}$$

For  $u = 1 - \cos t + \frac{e^{\frac{-x^2}{1+4t}}}{\sqrt{1+4t}}$

$$7) \begin{cases} u_t = u_{xx} \\ u|_{t=0} = xe^{-x^2} \end{cases} \Rightarrow u(t, x) = \frac{1}{2\sqrt{t}} \int_{-\infty}^{+\infty} e^{\frac{-(\xi-x)^2}{4t}} \xi e^{-\xi^2} d\xi =$$

$$= \frac{e^{\frac{-x^2}{1+4t}}}{2\sqrt{t}} \int_{-\infty}^{+\infty} \xi e^{-\left(\frac{\xi\sqrt{1+4t} - \frac{x}{\sqrt{1+4t}}}{2\sqrt{t}}\right)^2} d\xi \textcircled{=}$$

$$S = \xi \frac{\sqrt{1+4t}}{2\sqrt{t}} - \frac{x}{2\sqrt{t}\sqrt{1+4t}} \textcircled{=} \frac{e^{\frac{-x^2}{1+4t}}}{2\sqrt{t}} \int_{-\infty}^{+\infty} \left( \frac{2\sqrt{t}}{\sqrt{1+4t}} S e^{-S^2} + \frac{x}{1+4t} e^{-S^2} \right) dS$$

$$ds = \frac{\sqrt{1+4t}}{2\sqrt{t}} d\xi$$

$$= \frac{e^{\frac{-x^2}{1+4t}}}{2\sqrt{t}} \frac{x}{1+4t} \frac{2\sqrt{t}}{\sqrt{1+4t}} \int_{-\infty}^{+\infty} e^{-s^2} ds = \frac{x}{(1+4t)^{3/2}} \cdot e^{\frac{-x^2}{1+4t}}$$

13.6. 3)  $\begin{cases} u_t = \Delta u + \cos t \\ u|_{t=0} = xy e^{-x^2-y^2} \end{cases}$

$$u = \sin t + v \cdot w$$

$$\begin{cases} v_t = \Delta v \\ v|_{t=0} = x e^{-x^2} \end{cases} \quad \begin{cases} w_t = \Delta w \\ w|_{t=0} = y e^{-y^2} \end{cases}$$

$$u = \sin t + \frac{xy}{(1+4t)^3} e^{\frac{-(x^2+y^2)}{1+4t}}$$

~ ges. zsgen

## II. Aufgabe 2. Zweifelsfragen

$$\textcircled{1} \quad \begin{cases} \Delta u = 1, r < 2 \\ u|_{r=2} = 2\sin(\varphi + \frac{\pi}{3}) + 1 \end{cases}$$

$$u_r + \frac{1}{r}u_r + \frac{1}{r^2}u_{\varphi\varphi} = 1 \quad u_{\varphi\varphi} = \alpha r^2$$

$$2\alpha + 2\alpha + 0 = 1 \quad \alpha = \frac{1}{4} \quad u_{\varphi\varphi} = \frac{r^2}{4} \Rightarrow u = \frac{r^2}{4} + V, \quad \Delta V = 0$$

$$V|_{r=2} = (u - (\frac{r^2}{4})r)|_{r=2} = 2\sin(\varphi + \frac{\pi}{3}) + 1 - 1 = 2\sin(\varphi + \frac{\pi}{3}) =$$

$$= \sin\varphi + \sqrt{3}\cos\varphi; \quad \int V = A + B\sin\varphi + C\cos\varphi$$

$$\left. \begin{aligned} V|_{r=2} &= B\sin\varphi + C\cos\varphi \end{aligned} \right\} \Rightarrow \begin{cases} B=1 \\ C=\sqrt{3} \end{cases}$$

$$V = A + r(\sin\varphi + \sqrt{3}\cos\varphi)$$

$$u = \frac{r^2}{4} + A + 2r\sin(\varphi + \frac{\pi}{3}) \quad \leftarrow \text{erhet.}$$

$$\textcircled{2} \quad \begin{cases} \Delta u = \frac{18}{r^3} \sin 2\varphi \\ u|_{r=1/2} = 28\sin^2\varphi \\ u|_{r=1} = 4\cos^2\varphi + 5 \end{cases}$$

$$u_r + \frac{1}{r}u_r + \frac{1}{r^2}u_{\varphi\varphi} = \frac{18}{r^3} \sin 2\varphi \quad u_{\varphi\varphi} = \alpha \frac{1}{r} \sin 2\varphi$$

$$\alpha(-1)(-2) \frac{1}{r^3} \sin 2\varphi + \frac{1}{2}\alpha(-1) \frac{1}{r^2} \sin 2\varphi + \frac{1}{r^2} \alpha \frac{1}{r} (-4) \sin 2\varphi = \frac{18}{r^3} \sin 2\varphi$$

$$\Rightarrow \alpha = -6 \Rightarrow u_{\varphi\varphi} = -6 \frac{\sin 2\varphi}{r}$$

$$u = \frac{-6\sin 2\varphi}{r} + V, \quad \Delta V = 0$$

$$V|_{r=1/2} = \left( u - \frac{-6\sin 2\varphi}{r^2} \right) \Big|_{r=1/2} = 14 - 4\cos 2\varphi - 24\sin 2\varphi \quad (1)$$

$$V|_{r=1} = \left( u - \frac{-6\sin 2\varphi}{r^2} \right) \Big|_{r=1} = 7 + 2\cos 2\varphi - 6\sin 2\varphi \quad (2)$$

$$V = A + B\ln r + Cr^2 \cos 2\varphi + D \frac{1}{r^2} \sin 2\varphi + Er^2 \sin 2\varphi + F \frac{1}{r^2} \sin 2\varphi$$

$$\begin{cases} V_r|_{r=12} = 2B + C \cos 2\varphi - 6D \cos 2\varphi + E \sin 2\varphi - 6F \sin 2\varphi = (1) \\ V_r|_{r=1} = B + 2C \cos 2\varphi - 2D \cos 2\varphi + 2E \sin 2\varphi - 2F \sin 2\varphi = (2) \end{cases} \Rightarrow$$

$$\Rightarrow B = 7, C = 2, D = 1, E = -\frac{8}{5}, F = \frac{7}{2}$$

$$\text{Other: } u = \frac{-6 \sin 2\varphi}{r} + A + 7 \ln r + 2r^2 \cos 2\varphi + \frac{1}{r} \cos 2\varphi + \frac{7}{5r^2} \sin 2\varphi$$

$$\textcircled{3} \begin{cases} \Delta u = 24x = 24r \cos \varphi \\ (2u + u_r)|_{r=1} = 20 \sin^3 \varphi \end{cases}$$

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\varphi\varphi} = 24r \cos \varphi \quad u_2 = \alpha r^3 \cos \varphi$$

$$3 \cdot 2\alpha r \cos \varphi + 3\alpha r \cos \varphi - \alpha r \cos \varphi = 24r \cos \varphi \Rightarrow \alpha = 3 \Rightarrow$$

$$\Rightarrow u_2 = 3r^3 \cos \varphi \Rightarrow$$

$$\Rightarrow u = 3r^3 \cos \varphi + V, \Delta V = 0$$

$$(2V + V_r)|_{r=1} = (2u + 6r^3 \cos \varphi + u_r - (3r^3 \cos \varphi)_r)|_{r=1} =$$

$$= 20 \sin^3 \varphi - 6 \cos \varphi - 9 \cos \varphi = 15 \sin \varphi - 5 \sin 3\varphi - 15 \cos \varphi \quad (1)$$

$$V = A + B \sin \varphi + C \cos \varphi + D r^3 \sin 3\varphi$$

$$(2V + V_r)|_{r=1} = 2A + 2B \sin \varphi + 2C \cos \varphi + 2D \sin 3\varphi + B \sin \varphi +$$

$$C \cos \varphi + 3D \sin 3\varphi = (1) \Rightarrow \begin{cases} A = 0 \\ B = 5 \\ C = -5 \\ D = -1 \end{cases}$$

$$\text{Other: } u = 3r^3 \cos \varphi + 5r(\sin \varphi + \cos \varphi) - r^3 \sin 3\varphi$$

$$\textcircled{4} \int du = \frac{9}{r^2} \cos 3\varphi$$

$$\begin{cases} (u - u_r)|_{r=1} = \cos 3\varphi \end{cases}$$

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\varphi\varphi} = -\frac{9}{r^2} \cos 3\varphi \quad u_{rr} = \cos 3\varphi$$

$$0 + 0 + \frac{1}{r^2} (-9r) \cos 3\varphi = -\frac{9}{r^2} \cos 3\varphi \Rightarrow u_r = \cos 3\varphi$$

$$u = \cos 3\varphi + V, \quad V = 0$$

$$(V - V_r)|_{r=1} = (u - u_r)|_{r=1} + (-\cos 3\varphi + (\cos 3\varphi)_r)|_{r=1} = \frac{3}{4} \cos \varphi - \frac{3}{4} \cos 3\varphi$$

$$V = A + B \frac{1}{r} \cos \varphi + C \frac{1}{r^3} \cos 3\varphi$$

$$(V - V_r)|_{r=1} = A + B \cos \varphi + C \cos 3\varphi + B \cos \varphi + 3C \cos 3\varphi \Rightarrow \begin{cases} A = 0 \\ B = 3/8 \\ C = -3/16 \end{cases}$$

$$\Rightarrow u = \cos 3\varphi + \frac{3}{8r} \cos \varphi - \frac{3}{16r^3} \cos 3\varphi$$

$$\textcircled{5} \int du = \frac{4(x-y)^2}{(x^2+y^2)^3}, \quad 1 < r < 2$$

$$\begin{cases} u_r|_{r=1} = \frac{2xy}{x^2+y^2} \end{cases} \quad u_r + u|_{r=2} = \frac{(x+y)^2}{8(x^2+y^2)}$$

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\varphi\varphi} = 4 \frac{(\cos \varphi - \sin \varphi)^2}{r^4} = \frac{4}{r^4} - \frac{4 \sin 2\varphi}{r^4}$$

$$u_{r1} = \frac{\alpha}{r^2} \Rightarrow \frac{\alpha(-2)(-3)}{r^4} + \frac{1}{r} \alpha \frac{(-2)}{r^3} + \frac{1}{r^2} \cdot 0 = \frac{4}{r^4} \Rightarrow u_{r1} = \frac{1}{r^2}$$

$$u_{r2} = \frac{\alpha}{r^2} \sin 2\varphi \Rightarrow 0 = -\frac{4 \sin 2\varphi}{r^4} - \text{первое слагаемое}$$

$$u = f(r) \sin 2\varphi$$

$$f'' \sin 2\varphi + \frac{1}{r} f' \sin 2\varphi + \frac{1}{r^2} f (-4 \sin 2\varphi) = -\frac{4 \sin 2\varphi}{r^4} \Rightarrow$$

$$\Rightarrow u_{r2} = \alpha u_r \frac{1}{r^2} \sin 2\varphi, \quad (u_{r2})_r = -\frac{4 \sin 2\varphi}{r^4} \Rightarrow \alpha = 1 \Rightarrow$$

$$\Rightarrow u_{22} = \ln r \frac{1}{r^2} \sin 2\varphi$$

$$u = \frac{1}{r^2} + \frac{1}{r^2} \ln r \sin 2\varphi + V, \Delta V = 0$$

$$V_r = u_r + \frac{2}{r^3} + \left( \frac{2}{r^3} \ln r - \frac{1}{r^3} \right) \sin 2\varphi$$

$$V_r|_{r=1} = 2 \sin \varphi \cos \varphi + 2 - \sin 2\varphi = 2$$

$$(V_r + V)|_{r=2} = \frac{1}{8} (\sin^2 \varphi + \cos^2 \varphi + 2 \sin \varphi \cos \varphi) + \frac{1}{4} + \frac{1}{4} \ln 2 \sin 2\varphi - \frac{1}{8} \sin 2\varphi - \frac{1}{4} - \frac{1}{4} \ln 2 \sin 2\varphi = \frac{1}{8}$$

$$V = A + B \ln r$$

$$V_r|_{r=1} = B = 2$$

$$(V_r + V)|_{r=2} = \frac{B}{2} + A + B \ln 2 = \frac{1}{8} \Rightarrow A = -\frac{7}{8} - 2 \ln 2$$

$$u = \frac{1}{r^2} + \frac{1}{r^2} \ln r \sin 2\varphi - \frac{7}{8} - 2 \ln 2 + 2 \ln r$$

$$\textcircled{6} \quad \begin{cases} \Delta u = y, & r < 2, \quad r = \sqrt{x^2 + y^2} \\ u_r|_{r=2} = \sin^3 \varphi + \alpha \cos^2 \varphi \end{cases}$$

$$\int_{r=2} (\sin^3 \varphi + \alpha \cos^2 \varphi) dS = \int_0^{2\pi} (\sin^3 \varphi + \alpha \cos^2 \varphi) 2 d\varphi$$

$$\int_0^{2\pi} (1 - \cos^2 \varphi) \sin \varphi d\varphi + \alpha \int_0^{2\pi} (1 + \cos 2\varphi) d\varphi =$$

$$= 2 \left( -\cos \varphi + \frac{\cos^3 \varphi}{3} \right) \Big|_0^{2\pi} + \alpha \left( \varphi + \frac{\sin 2\varphi}{2} \right) \Big|_0^{2\pi} = 2\alpha\pi$$

$$\iint_{r < 2} y dx dy = \int_0^{2\pi} d\varphi \int_0^2 r \sin \varphi r dr = 0 \quad \Rightarrow \underline{\alpha = 0}$$

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\varphi\varphi} = r \sin \varphi, \quad u_2 = \alpha r^3 \sin \varphi$$

$$6\alpha r \sin\varphi + 3\alpha r \sin\varphi - \alpha r \sin\varphi = 2 \sin\varphi \Rightarrow \alpha = \frac{1}{8}$$

$$u = \frac{1}{8} r^3 \sin\varphi + V \quad V_r|_{r=2} = \sin^3\varphi - \frac{3}{2} \sin\varphi$$

$$V = A r \sin\varphi + B r^3 \sin 3\varphi$$

$$\frac{3}{4} \sin\varphi - \frac{1}{4} \sin 3\varphi$$

$$V_r|_{r=2} = A \sin\varphi + 12B \sin 3\varphi = -\frac{3}{4} \sin\varphi - \frac{1}{4} \sin 3\varphi \Rightarrow \begin{cases} A = -3/4 \\ B = -1/48 \end{cases}$$

$$u = \frac{1}{8} r^3 \sin\varphi - \frac{3}{4} r \sin\varphi - \frac{1}{48} r^3 \sin 3\varphi$$

$$\textcircled{7} \text{ a) } Au = \frac{1}{r^4}, \quad r > 2$$

$$(u - u_r)|_{r=2} = \sin\theta \cos^2\frac{\theta}{2} \sin\left(\frac{\pi}{6} - \varphi\right)$$

$$u_{rr} + \frac{2}{r} u_r + \dots = \frac{1}{r^4} \Rightarrow u_{\text{outer}} = \frac{\alpha}{r^2}$$

$$\frac{\alpha(-2)(-3)}{r^4} + \frac{2}{r} \frac{\alpha(-2)}{r^3} = \frac{1}{r^4} \Rightarrow \alpha = 1/2$$

$$u = \frac{1}{2r^2} + V, \quad (V - V_r)|_{r=2} = (u - u_r)|_{r=2} - \frac{1}{8} - \frac{1}{8} =$$

$$\frac{1}{2} \sin\theta (1 + \cos\theta) \left( \frac{1}{2} \cos\varphi - \frac{\sqrt{3}}{2} \sin\varphi \right) - \frac{1}{4} =$$

$$= \underbrace{\frac{1}{2} \sin\theta \left( \frac{1}{2} \cos\varphi - \frac{\sqrt{3}}{2} \sin\varphi \right)}_{J_1^*} + \underbrace{\frac{1}{2} \sin\theta \cos\theta \left( \frac{1}{2} \cos\varphi - \frac{\sqrt{3}}{2} \sin\varphi \right)}_{J_2^*} - \underbrace{\frac{1}{4}}_{J_0^*}$$

$$V = \frac{1}{r} J_0 + \frac{1}{r^2} J_1 + \frac{1}{r^3} J_2$$

$$(V - V_r)|_{r=2} = \frac{1}{2} J_0 + \frac{1}{4} J_1 + \frac{1}{8} J_2 + \frac{1}{4} J_0 + \frac{2}{8} J_1 + \frac{3}{16} J_2 = J_0^* + J_1^* + J_2^* \Rightarrow$$

$$\Rightarrow \begin{cases} J_0 = \frac{4}{3} J_0^* = -\frac{1}{3} \\ J_1 = 2 J_1^* = \sin\theta \sin\left(\frac{\pi}{6} - \varphi\right) \\ J_2 = \frac{16}{5} J_2^* = \frac{8}{5} \sin\theta \cos\theta \sin\left(\frac{\pi}{6} - \varphi\right) \end{cases}$$

$$u = \frac{1}{2r^2} - \frac{1}{3r} + \frac{1}{r^2} \sin \theta \sin\left(\frac{\pi}{6} - \varphi\right) + \frac{1}{r^3} \cdot \frac{8}{5} \sin \theta \cos \theta \sin\left(\frac{\pi}{6} - \varphi\right)$$

6.26. 3)  $\Delta u = 0, r < 1$

$$u|_{r=1} = \sin \theta (\sin \varphi + \sin \theta) = \overbrace{\sin \theta \sin \varphi}^{J_1^*} + \sin^2 \theta$$

$$\sin^2 \theta = 1 - \cos^2 \theta = - \underbrace{\frac{2}{3} \cdot \frac{1}{2} (3 \cos^2 \theta - 1)}_{J_2^*} + \underbrace{\frac{2}{3}}_{J_0^*}$$

$$u|_{r=1} = J_0^* + J_1^* + J_2^*$$

$$u = J_0 + r J_1 + r^2 J_2$$

$$u|_{r=1} = J_0 + J_1 + J_2 = J_0^* + J_1^* + J_2^* \Rightarrow \begin{cases} J_0 = 2/3 \\ J_1 = \sin \theta \sin \varphi \\ J_2 = -\frac{1}{3} (3 \cos^2 \theta - 1) \end{cases}$$

$$u = \frac{2}{3} + r \sin \theta \sin \varphi + r^2 \left(-\frac{1}{3}\right) (3 \cos^2 \theta - 1)$$

6.31. 1)  $\Delta u = 0, 1 < r < 2$

$$(3u + ur)|_{r=1} = \underbrace{5 \sin^2 \theta \sin^2 \varphi}_{J_2^*}, \quad u|_{r=2} = - \underbrace{\cos \theta}_{J_1^*}$$

$$u = r^2 J_2 + \frac{1}{r^3} \tilde{J}_2 + r J_1 + \frac{1}{r^2} \tilde{J}_1 - \text{гмт. условия на } r=2$$

$$\begin{cases} (3u + ur)|_{r=1} = 3J_2 + 3\tilde{J}_2 + 3J_1 + 3\tilde{J}_1 + 2J_2 - 3\tilde{J}_2 + J_1 - 2\tilde{J}_1 = J_2^* \\ u|_{r=2} = 4J_2 + \frac{1}{8}\tilde{J}_2 + 2J_1 + \frac{1}{4}\tilde{J}_1 = J_1^* \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} J_2 = \frac{1}{5} J_2^* \\ \tilde{J}_2 = -\frac{32}{5} J_2^* \end{cases} \quad \begin{cases} J_1 = J_1^* \\ \tilde{J}_1 = -4 J_1^* \end{cases}$$

6.27. 3)  $\Delta u = 0, r < R$

$$u|_{r=R} = \sin^2 \theta \cos\left(2\varphi - \frac{\pi}{4}\right) + \sin \theta \sin \varphi$$

$$J_2^* \leftarrow \sin^2 \theta \frac{1}{\sqrt{2}} (\cos 2\varphi + \sin 2\varphi) \quad J_1^*$$

$$u = rJ_1 + r^2 J_2$$

$$u|_{r=R} = RJ_1 + R^2 J_2 = J_1^* + J_2^* \Rightarrow \begin{cases} J_1 = \frac{J_1^*}{R} \\ J_2 = \frac{J_2^*}{R^2} \end{cases}$$

$$u = \frac{r}{R} \sin \theta \sin \varphi + \frac{r^2}{R^2} \sin^2 \theta \cos(\varphi - \pi/4)$$

16.28. 2)  $\int \Delta u = 0, r > 1$

$$u|_{r=1} = \cos^2 \theta \sin \theta \sin(\varphi + \frac{\pi}{3}) = \cos^2 \theta \sin \theta \left( \frac{1}{2} \sin \varphi + \frac{\sqrt{3}}{2} \cos \varphi \right)$$

$$P_3'(\cos \theta) = -\frac{2}{5} \sin \theta (5 \cos^2 \theta - 1)$$

$$P_1'(\cos \theta) = -\sin \theta$$

$$\cos^2 \theta \sin \theta = -\frac{2}{5} P_3'(\cos \theta) - \frac{1}{5} P_1'(\cos \theta)$$

$$u|_{r=1} = \alpha P_3'(\cos \theta) \sin \varphi + \beta P_1'(\cos \theta) \cos \varphi + \gamma P_3'(\cos \theta) \sin \varphi + \delta P_1'(\cos \theta) \cos \varphi = \left( -\frac{2}{5} P_3'(\cos \theta) - \frac{1}{5} P_1'(\cos \theta) \right) \left( \frac{1}{2} \sin \varphi + \frac{\sqrt{3}}{2} \cos \varphi \right) \Rightarrow$$

$$\Rightarrow \alpha = -\frac{1}{10}, \beta = \frac{\sqrt{3}}{10}, \gamma = -\frac{1}{15}, \delta = -\frac{\sqrt{3}}{15}$$

$$u = -\frac{\sqrt{3}}{10} \frac{1}{r^2} P_1' \cos \varphi - \frac{1}{10} \frac{1}{r^2} P_1' \sin \varphi - \frac{\sqrt{3}}{15} \frac{1}{r^4} P_3' \cos \varphi - \frac{1}{15} \frac{1}{r^4} P_3' \sin \varphi$$

$$u = \left( \frac{1}{5r^2} \sin \theta + \frac{5 \cos^2 \theta - 1}{5r^4} \sin \theta \right) \sin(\varphi + \frac{\pi}{3})$$